
REPORTNOW

RISK MANAGEMENT PLAN

Version 1.0

12/10/2025

VERSION HISTORY

Version #	Implemented By	Revision Date	Approved By	Approval Date	Reason
1.0	Madelyn Cruz Tan	12/10/2025	Tan Yong Huat	12/10/2025	Initial Risk Management Plan draft

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1 INTRODUCTION

1.1 PURPOSE OF THE RISK MANAGEMENT PLAN

A Risk Management Plan ensures that potential uncertainties are systematically anticipated, evaluated, and addressed before they can affect project outcomes. In the case of ReportNow, these risks may include delays in AI integration, user privacy breaches, or complications in coordinating across multiple government agencies. The plan defines the framework for identifying, analysing, mitigating, and tracking such risks throughout the project's lifecycle, embedding risk awareness into every stage of development.

Beyond outlining mitigation strategies, the plan also specifies how risks will be documented, prioritised, and continuously monitored. This helps the team respond to emerging issues in a structured and timely way rather than reactively. Developed by the ReportNow project manager during the Planning Phase and refined as the project advances, this document serves as a guide for the project team and university supervisors in maintaining oversight and ensuring alignment with project milestones.

1.2 SCOPE AND OBJECTIVES OF RISK MANAGEMENT PLAN

This Risk Management Plan applies to all stages of the *ReportNow* project, from initial design and prototyping to deployment and maintenance. It focuses on identifying, analysing, and mitigating risks that may impact the project's scope, schedule, cost, or quality. The objectives are to minimise potential disruptions, maintain compliance with security and data protection standards, and ensure smooth collaboration among team members and external stakeholders.

1.3 COMMON RISK CATEGORIES

Several categories of risk are relevant to ReportNow's development:

- **Resource Risks:** Insufficient manpower, delays in deliverables, or limited access to testing environments may hinder progress.
- **Software Risks:** Bugs, integration errors, or design flaws could affect platform stability and user experience.
- **Communication Risks:** Misaligned expectations or unclear documentation might result in duplicated effort or missed dependencies.
- **External Risks:** Dependencies on government APIs, changing policy requirements, or third-party service outages could disrupt operations.
- **Hardware Risks:** Failures in testing devices or server infrastructure may lead to downtime during key development phases.

2 RISK MANAGEMENT PROCEDURE

2.1 PROCESS

The ReportNow project manager, together with the development team and project sponsors, is responsible for ensuring that risks are actively identified, assessed, and managed throughout all phases of the project lifecycle. Risk awareness begins early in planning and continues through implementation, testing, and deployment to minimise potential disruption.

Risks will be documented and reviewed regularly, with preventive measures or contingency plans developed where appropriate. The project will maintain an open communication channel for reporting emerging risks, allowing prompt evaluation and response.

The ReportNow project manager will assume the role of Risk Manager, coordinating with team leads to ensure that each risk is logged, prioritised, and tracked until resolution. Detailed steps and responsibilities for this process are described in the subsequent sections.

2.2 RISK IDENTIFICATION

Risk identification for ReportNow will be a collaborative and ongoing effort involving the project manager, development team, and relevant stakeholders. The process will include evaluating environmental factors, organisational culture, and the overall project management plan — including scope, assumptions, and constraints. Key project artefacts such as deliverables, the Work Breakdown Structure (WBS), cost and effort estimates, and the resource plan will be examined to surface potential risks early and reduce their impact on the project's objectives.

All identified risks will be recorded in the Risk Management Log, which will be maintained electronically and updated as the project progresses. The log will serve as the central repository for tracking and analysing risks, ensuring transparency and accountability across the team. It can be accessed via the shared project library at:

 TEL5_Team2_Risk_Management_Log.xlsx

2.3 RISK ANALYSIS

All risks identified for ReportNow will undergo structured analysis to determine their potential effect on project objectives and outcomes. Each risk will first be qualified through an assessment of its likelihood and impact, allowing the team to distinguish between minor issues that can be accepted and significant risks that demand proactive mitigation.

This analysis process helps the team focus resources on high-priority risks that could disrupt core aspects of ReportNow, such as system reliability, data security, or integration with government APIs. Quantitative measures such as probability–impact ratings will be applied where appropriate to provide a clearer understanding of overall project exposure.

The results of each analysis will be reflected in the Impact–Probability Matrix and updated in the Risk Management Log, ensuring that prioritisation remains transparent and actionable throughout the project lifecycle.

2.3.1 QUALITATIVE RISK ANALYSIS

The probability and impact of each identified risk will be evaluated by the ReportNow project manager, with input from the development team and relevant stakeholders. This assessment provides a structured way to gauge the severity of potential issues and to prioritise which ones require active response planning.

Probability Levels

- High: Greater than 70% chance of occurrence.
- Medium: Between 30% and 70% chance of occurrence.
- Low: Below 30% chance of occurrence.

Impact Levels

- High: Risk has the potential to significantly affect project cost, schedule, or performance.
- Medium: Risk may cause moderate delays or resource strain but remains manageable with mitigation.
- Low: Risk is expected to have minimal or negligible effect on project objectives.

These two dimensions—probability and impact—will be plotted on an Impact–Probability Matrix to visualise and prioritise risks. Risks appearing in the red and yellow zones of the matrix will trigger the development of targeted response plans, which may include both mitigation strategies and contingency actions to reduce their likelihood or effect. The table below shows an example of an Impact–Probability Matrix to be included in our Risk Management Log.

Impact	High	1. Hardware failure e.g., Hard Drive corrupted 2. System not meeting requirements 3. Misunderstanding/Miscommunication of requirements from stakeholders 4. Sensitive data being leaked/hacked 5. User impersonation or spam reports 6. Non-compliance with PDPA or Smart Nation security guidelines	1. Unable to meet Project Deadline 2. Feature creep 3. Changes in government data policy or API access 4. Inadequate test coverage or late defect discovery	
	Medium	1. Poor code design (e.g. high coupling) 2. Inconsistent Programming Style	1. Staff Unavailability 2. AI misclassifies or fails to detect duplicate reports 3. Inaccurate cost or schedule estimation	1. Updates to code not communicated to every group member
	Low	1. Incomplete Documentation		
		Low	Medium	High
Probability				

Figure 1: Impact-Probability Matrix

2.3.2 QUANTITATIVE RISK ANALYSIS

For risks prioritised through the qualitative analysis, further evaluation will be conducted to quantify their potential impact on ReportNow's cost, schedule, and overall performance. This process assigns measurable values to the probability and effect of each major risk, providing a more objective view of how they may influence project outcomes.

Each prioritised risk will receive a numerical rating derived from probability-impact scoring or scenario-based estimation, helping the team compare and rank risks based on total exposure. Where feasible, these analyses may include projected cost variances, potential schedule delays, or performance degradation metrics.

The resulting quantitative data will be documented in the Risk Management Log and used to refine mitigation priorities, validate contingency budgets, and support informed decision-making throughout the ReportNow project lifecycle.

2.4 RISK RESPONSE PLANNING

Major risks, appearing in the red or yellow zones of the Impact–Probability Matrix will be assigned to a specific team member for active monitoring. This ensures accountability and prevents any significant risk from being overlooked

For every high-priority risk, the project team will determine an appropriate response strategy from one of the following approaches:

- Avoid: Eliminate the threat entirely by removing its root cause.
- Mitigate: Reduce the probability or impact of the risk through preventive measures or design adjustments.
- Accept: Acknowledge the risk and take no immediate action beyond monitoring.
- Transfer: Shift responsibility to another party, such as through outsourcing or insurance coverage.

When a mitigation strategy is chosen, the team will identify concrete actions to reduce the risk's likelihood or potential impact. This may involve early prototyping, revising the project schedule, allocating additional resources, or enhancing testing and validation processes.

For risks that are mitigated or accepted, a predefined contingency plan will be developed to outline the steps to take should the risk materialise. This ensures that, even in adverse scenarios, ReportNow can continue operations with minimal disruption and maintain alignment with its project objectives.

2.5 RISK MONITORING, CONTROLLING, AND REPORTING

The ReportNow project's risk profile will be tracked and reviewed throughout its lifecycle to ensure timely detection of new risks and effective management of existing ones.

A Top 10 Risk List will be maintained as part of routine project status reports, highlighting high-priority risks and monitoring the progress of mitigation efforts.

All change requests will be assessed for potential impact on current risks or the creation of new ones. Significant updates to risk status will be communicated to management and project sponsors through the Executive Project Status Report for prompt action.

3 TOOLS AND PRACTICES

A Risk Management Log will be maintained by the ReportNow QA team and project manager and reviewed regularly as a standing item in project team meetings. This ensures that risk awareness remains active, updates are promptly recorded, and mitigation actions are consistently tracked.

Each log entry will include

- Version Control Details: Tracks revisions to the risk entry and ensures all updates or changes are properly documented over time.
- Unique Risk Identifier: Assigns a distinct reference number to each risk for easy tracking and cross-referencing in reports.
- Risk Type: Categorises the risk (e.g., technical, resource, communication, external) to support pattern analysis and response planning.
- Description: Provides a clear summary of the nature and cause of the identified risk.
- Probability: Estimates the likelihood of the risk occurring, typically rated on a qualitative or numerical scale.
- Impact: Describes the potential consequences on project objectives such as scope, cost, schedule, or quality.
- Numerical Rating: Combines probability and impact scores to quantify overall risk severity and guide prioritisation.
- Potential Ramifications: Outlines the possible downstream effects if the risk is realised, including secondary impacts or dependencies.
- Team Member Responsible: Identifies the individual accountable for monitoring and managing the specific risk.
- Proposed Approach: Summarises the intended strategy for handling the risk—avoidance, mitigation, transfer, or acceptance.
- Impact–Probability Matrix: A visual tool used to map risks by their likelihood and impact levels, helping the team quickly distinguish between low-priority and high-priority risks.
- Top 10 Risks: A dynamic list spotlighting the most critical risks requiring regular review and immediate mitigation attention.

RISK MANAGEMENT PLAN APPROVAL

The undersigned acknowledge they have reviewed the **Risk Management Plan** for the ReportNow project. Changes to this Risk Management Plan will be coordinated with and approved by the undersigned or their designated representatives.

Signature:  Date: 12/10/2025

Print Name: Tan Yong Huat

Role: QA Manager

Signature:  Date: 12/10/2025

Print Name: Choo Kean Yee

Role: QA Engineer

Signature:  Date: 12/10/2025

Print Name: Madelyn Cruz Tan

Role: Project Manager

APPENDIX A: REFERENCES

The following table summarizes the documents referenced in this document.

Document Name and Version	Description	Location
Risk Management Log Version 1.0	An excel workbook containing the list of risks and risk analysis performed.	 TEL5_Team2_Risk_Man...
ReportNow Software Quality Assurance (SQA) Plan Version 1.00	The Software Quality Assurance (SQA) Plan establishes the goals, processes, and responsibilities required to implement effective quality assurance functions for the ReportNow project.	 TEL5_Team2_Quality_P...
CDC UP Risk Management Plan Template	ReportNow Risk Management Plan is based on the CDC UP Risk Management Plan Template.	NTULearn Blackboard

APPENDIX B: KEY TERMS

The following table provides definitions for terms relevant to the Risk Management Plan.

Term	Definition
Quality Assurance (QA)	The process of evaluating overall project performance on a regular basis to ensure that it will satisfy the relevant quality standards.
Work Breakdown Structure (WBS)	WBS in project management is a deliverable-oriented breakdown of a project into smaller components.